

Document Number: P3576R0
Date: 2025-1-13
Authors: Michael Wong
Project: Programming Language C++, SG19 Machine Learning
Reply to: Michael Wong <fraggamuffin@gmail.com>

SG19: Machine Learning virtual Meeting Minutes to 2024/11/14-2024/12/12

Contents

Minutes for 2024/11/14 SG19 Conference Call.....	1
Minutes for 2024/12/12 SG19 Conference Call.....	15

Minutes for 2024/11/14 SG19 Conference Call

Notes: On Thu, Nov 14, 2024 at 7:56 AM Michael Wong <fraggamuffin_at_[hidden]> wrote:

- > Hi I am not sure if this means we should hold a meeting. Given that Oliver
- > just wrote a D paper with Mark,it would be good to have a look at it. So
- > even though it will be 3 am, I will set it up. I will ask Phil to chair,and
- > will wake up to start it, though I will likely not be able to stay up
- > through it. Thank you.
- >
- > Hi, this SG19 meeting will focus on stats and graphs.
- >
- > Michael Wong is inviting you to a scheduled Zoom meeting.
- >
- > Topic: SG19 monthly
- > Time: 2nd Thursdays 02:00 PM Eastern Time (US and Canada)
- > Every month on the Second Thu,
- >
- >
- > Join from PC, Mac, Linux, iOS or Android:
- >
- > <https://iso.zoom.us/j/93084591725?pwd=K3QxZjJlcnljaE13ZWU5cTILNkx0Zz09>
- > Password: 035530

- >
- > Or iPhone one-tap :
- > US: +13017158592,,93084591725# or +13126266799,,93084591725#
- > Or Telephone:
- > Dial(for higher quality, dial a number based on your current location):
- > US: +1 301 715 8592 or +1 312 626 6799 or +1 346 248 7799 or +1
- > 408 638 0968 or +1 646 876 9923 or +1 669 900 6833 or +1 253 215 8782
- > or 877 853 5247 (Toll Free)
- > Meeting ID: 930 8459 1725
- > Password: 035530
- > International numbers available: <https://iso.zoom.us/j/93084591725>
- >
- > Or Skype for Business (Lync):
- > <https://iso.zoom.us/j/93084591725>
- >
- > Agenda:
- >
- > 1. Opening and introductions
- >
- > The ISO Code of conduct:
- > <https://www.iso.org/files/live/sites/isoorg/files/store/en/PUB100397.pdf>
- >
- > IEC Code of Conduct:
- >
- > <https://www.iec.ch/basecamp/iec-code-conduct-technical-work>
- >
- > ISO patent policy.
- >
- >
- >
- > https://isotc.iso.org/livelink/livelink/fetch/2000/2122/3770791/Common_Policy.htm?no_deid=6344764&vernum=-2
- >
- > The WG21 Practices and Procedures and Code of Conduct:
- >
- > <https://isocpp.org/std/standing-documents/sd-4-wg21-practices-and-procedures>
- >
- > 1.1 Roll call of participants
- >
- > Oliver, Richard, Pete, Mark, Michael, Jens, Scott

> 1.2 Adopt agenda
>
> 1.3 Approve minutes from previous meeting, and approve publishing
> previously approved minutes to ISOCPP.org
>
> 1.4 Action items from previous meetings
>
> 2. Main issues (125 min)
>
> 2.1 General logistics
>
>
>
> * Jan 11, 2024 02:00 PM ET: Graph DONE
> * Feb 8, 2024 02:00 PM ET: Graph DONE
> * Mar 14, 2024 02:00 PM ET: Cancelled due to Tokyo 3-18-23
> * Apr 11, 2024 02:00 PM ET: Stats/Graph DONE
> * May 9, 2024 02:00 PM ET: Graph DONE
> * June 13, 2024 02:00 PM ET: Graph; St.louis 6-24-29 DONE
> * July 11, 2024 02:00 PM ET: Stats/ Graphs DONE
> * Aug 15, 2024 02:00 PM ET: Graph DONE
> * Sep 12, 2024 02:00 PM ET: CPPCON Sept 15-20 so canceled DONE
> * Oct 10, 2024 02:00 PM ET: Stats
> * Nov 14, 2024 02:00 PM ET: Wroclaw F2F
> * Dec 12, 2024 02:00 PM ET: Graph
>
>
> ISO meeting status
>
> future C++ Std meetings
>
> 2.2 Paper reviews
>
>
RESTART AT 2:45 PM Eastern TIME

P3495 from Oliver/Mark

>

P3495R0.pdf <<https://isocpp.org/files/papers/P3495R0.pdf>>

>From Mark: for ``vector_two_norm``:

<https://eel.is/c++draft/linalg#algs.blas1.nrm2-3>

Returns: The square root of the sum of the square of init and the squares of the absolute values of the elements of v.

Ask Inbal: Why isn't my paper on the agenda.

<https://github.com/orgs/cplusplus/projects/30>

Sg19 stats Oliver feedback

The discussion focused on the SG19 stats paper, highlighting issues with unspecified values, state-of-the-art design, and API concerns. Key points included the need to address unspecified results due to non-finite inputs, insufficient elements, and floating-point exceptions. The boost accumulators' performance charts lack error bars, and their parallel execution should be explored. Accuracy trade-offs and the need for consistent results between free functions and accumulators were discussed. API concerns were raised about defaulted Booleans, return types, and the potential for a constructor taking a range of elements. Explicit template parameters for initial values were proposed to improve clarity.

Action Items

- [] Expand the discussion on unspecified values, including justification for the handling of NaNs, infinities, and insufficient elements.
- [] Provide more details on the Boost accumulators, including performance data with error bars.
- [] Explore the design space around handling ranges with insufficient elements in depth, and justify the choices made in the paper.
- [] Discuss the accuracy and parallel processing aspects of the design, including guarantees provided by the free functions versus the accumulators.
- [] Address the API concerns, such as the use of defaulted Booleans and the return type of the accumulator's operator().
- [] Consider the proposal to allow explicitly specifying an initial value for the accumulators.

Unspecified Values in the Paper

- discusses the issue of unspecified values in the paper, highlighting the large surface area of unspecified behavior.
- Concerns are raised about the implications of unspecified values at run-time and compile-time.
- questions whether the C standard library functions' behavior regarding floating point exceptions should be carried over to statistical functions and accumulators.
- The paper states that a statistic is unspecified if the range is consumed by nones or infinities, or if the range has an insufficient number of elements.

Handling Infinities and Nans

- questions the justification for yielding unspecified results when inputting infinities or Nans.
- The C standard precisely specifies what happens in these situations, and WG 21 has expressed a preference for Nans and infinities down.
- raises the issue of whether Fe invalid should be raised when feeding in infinities or Nans.
- The discussion includes the case of insufficient elements and the need for a broader design space discussion.

State of the Art and Boost Accumulators

- mentions the broken link in the boost accumulators and the need for more detail for readers of 1708.
- The performance charts could be improved by including error bars or stating if errors are too small to be metricable.
- notes that the merged accumulator is inferior for computing the mean and variance and suggests exploring this further.
- The discussion includes the need for a broader discussion about the design space and trade-offs in 1708.

Accuracy and Sorting Ranges

- discusses the trade-off between accuracy and speed, suggesting sorting ranges from smallest to largest magnitude before feeding to statistical functions.
- There are no guarantees on how statistical functions would process sorted ranges, and this needs to be specified better.
- The question arises whether free functions and accumulators should

give the same results, and the current reference implementation does not satisfy this property.

- The discussion includes the need to explore how accumulators work with parallel acceleration and the trade-offs involved.

API Concerns and Design Space Considerations

- raises concerns about defaulted Booleans and return types, suggesting more expressive alternatives.

- proposes using designated initializations to make the code more reasonable.

- suggests that accumulators should support a constructor that takes a range or a single element.

- The discussion includes the need to explicitly specify initial values for templates and the potential benefits of this approach.

>From Phil: There are updates to the P3126 Overview and P3127 Background and

> Terminology papers for the Graph Library. They haven't been published yet,

> but it would be helpful to get input on them before they're published.

>

>

>

> Since the changes are a response to Oliver's concerns, I think Oliver and

> Andrew would need to attend to make it useful to have a session for the

> Graph Library. Unless they want to do that, I think we can send draft

> versions to the reflector, unless there's a better idea.

>

>

> Review BSI Graph feedback:

> As Oliver (Rosten) said "The basic premise is important, and it would be

> fantastic to have support for graphs in the standard."

>

> The main items identified were:

> Oliver:

> - This paper is long and incomplete, it has lots of details which I think

> to be irrelevant, however things that are definitely relevant are missing

> from the paper - for example definition of graph - since people have

> different ideas. We need to add a mathematical perspective to the paper.

>

- > - The structure of the paper completely changed in the new revision, so now
- > it's hard to understand what and why they have done
- >
- > - Another missing part is discussion of graph invariants
- >
- > Tom (Deakin): There's a big missing part in "Prior art" part, GraphBLAS (<https://graphblas.org>) eminently.
- >
- > Some other things to add:
- >
- > 1. The electrical circuit example needs more explanation, and I think this
- > will highlight some deep issues around representing things which are
- > seemingly trivially graphs, as graphs in practice. In what sense is a
- > bog-standard resistor directed? I assume the reason that the graph is
- > directed is because current has a sign and in an undirected graph it
- > becomes ambiguous which way the current is flowing (also you may want
- > components like diodes). But the directed representation also has issues:
- > "can current flow from 'Vdd' to 'n0'?" should be immediately answerable
- > from the properties of Vdd and its edges. There are other ways to represent
- > an electrical circuit. One is as a directed graph but with incident edges
- > recorded - but iirc, this is excluded from the latest version of the paper.
- > Alternatively, one could have a mathematical object, the name of which I
- > actually don't know: it looks like an undirected graph, but where each
- > partial edge has additional, unique, end-point data, as well as the common
- > weight. Things like this are the reason why I think we need a broader group
- > to look at this proposal (i.e. beyond SG19) and if we possibly can we
- > should involve someone from the mathematics community. Otherwise there's a
- > real danger we end up missing important insights.
- >
- > 2. My comment about the structure of the paper changing was a reference to
- > previous comparisons with boost::graph. I'm sure these were in an earlier
- > version, or am I misremembering? Either way, it would be very helpful to
- > have a proper discussion of e.g. the move away from visitors.
- >
- > 3. Re. the definition of a graph, there needs to be a proper discussion
- > about whether the paper's definition of graph is what some authors call a
- > multigraph and whether it does/does not include loops. These things are
- > mentioned, in passing, when introducing algorithms, but terminology needs
- > to be properly established.
- >
- > 4. I think we're trying to do too much in one go in this paper. I think a

> great first step would be to build on mdspan and try to standardize (or at
> least understand) what might reasonably be called an unstructured span.
> This could be represented as a vector of vectors or as a vector with some
> auxiliary storage indicating where the partitions fall. The point is that
> an unstructured span, with the right invariants, is an adjacency list. If
> we can understand unstructured span and its desirable api, I think this
> will be incredibly valuable guidance for what a standardized graph
> container might look like.

>

> 5. IIUC, this paper excludes pure connectivity graphs. These are incredibly
> helpful and, if I've understood correctly that they are not supported,
> would be a major omission. Another good reason, imo, to start with
> unstructured span!

>

> 6. I'm not convinced by the load api. We don't have a load api for vector
> etc. Moreover, would it not be preferable to have appropriate constructors?

>

>

> 2.2.1: ML topics

>

> 2.2.1.1 Graph Proposal Phil Ratsloff et al

>

> Latest paper:

>

> Here's a link to the paper (different than the previous paper reviewed).

> There are some additional updates I'm planning on making before the
> meeting.

>

>

>

<https://docs.google.com/document/d/1OpH-xxRri7tJTtJJIZTYmSHkkrZJkdBwm9zJ7LqolfQ/edit?usp=sharing>

>

>

>

>

> P1709R3:

>

>

https://docs.google.com/document/d/1kLHhbSTX7j0tPeTYECQFSNx3R35Mu3xO5_dyYdRy4dM/edit?usp=sharing

>
>
>
https://docs.google.com/document/d/1QkfDzGyfNQKs86y053M0YHOLP6frzhTJqzg1Ug_vkkE/edit?usp=sharing
>
> <<http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2020/p2119r0.html>>
>
> <
>
>
<https://docs.google.com/document/d/175wlm8o4BNGti0WLq8U6uZORegKVjmnpc-E8PoGS0/edit?ts=5fff27cd#heading=h.9ogkehmdmtel>
> *>*<
>
> Array copy semantics:
> array copy-semantics paper P1997 "Relaxing Restrictions on Arrays",
> <https://wg21.link/p1997>
>
> Stats feedback:
>
> P2376R0
> <<http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2021/p2376r0.pdf>>
> Comments
> on Simple Statistical Functions (p1708r4): Contracts, Exceptions and
> Special cases Johan Lundberg
>
> 2.2.1.2 Reinforcement Learning Larry Lewis Jorge Silva
>
> Reinforcement Learning proposal:
>
> 2.2.1.3 Differential Calculus:
>
>
>
<https://docs.google.com/document/d/175wlm8o4BNGti0WLq8U6uZORegKVjmnpc-E8PoGS0/edit?ts=5fff27cd#heading=h.9ogkehmdmtel>
>
> 2.2.1.4: Stats paper
>
> P2681R0

- > <<https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2022/p2681r0.pdf>>
- > More
- > Stats Functions Richard Dosselmann, Michael Wong
- > Current github
- >
- > <https://github.com/cplusplus/papers/issues/475>
- >
- > <https://github.com/cplusplus/papers/issues/979>
- >
- > Stats review Richard Dosselman et al
- >
- > <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2021/p1708r4.pdf>
- >
- > Feedback from Johan Lundberg and Oleksandr Korval
- >
- > <https://isocpp.org/files/papers/D2376R0.pdf>
- >
- > P1708R3: Math proposal for Machine Learning: 3rd review
- >
- > PXXXX: combinatorics: 1st Review
- >
- > *> std.org/jtc1/sc22/wg21/docs/papers/2020/p1708r2
- > <<http://std.org/jtc1/sc22/wg21/docs/papers/2020/p1708r2>>*
- > *> above is the stats paper that was reviewed in Prague*
- > *> <http://wiki.edg.com/bin/view/Wg21prague/P1708R2SG19>
- > <<http://wiki.edg.com/bin/view/Wg21prague/P1708R2SG19>>*
- > *>*
- > *> Review Jolanta Polish feedback.*
- > *> <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2020/p2119r0.html>
- > <<http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2020/p2119r0.html>>*
- >
- >
- > 2.2.1.4: Matrix paper
- >
- > 2.2.3 any other proposal for reviews?
- >
- > 2.3 Other Papers and proposals
- >
- > P1416R1: SG19 - Linear Algebra for Data Science and Machine Learning
- >
- >

<https://docs.google.com/document/d/1IKUNiUhBgRURW-UkspK7fAAylhfXuMxjk7xKikK4Yp8/edit#heading=h.tj9hitg7dbtr>

>
> P1415: Machine Learning Layered list

>
>
https://docs.google.com/document/d/1eINFdIXWoetbxjO1OKol_Wj8fyi4Z4hogfj5tLVSi64/edit#heading=h.tj9hitg7dbtr

>
> 2.2.2 SG14 Linear Algebra progress:
> Different layers of proposal

>
>
https://docs.google.com/document/d/1poXfr7mUPovJC9ZQ5SDVM_1Nb6oYAXIK_d0ljdUAtSQ/edit

>
> 2.5 Future F2F meetings:
>
> 2.6 future C++ Standard meetings:
> <https://isocpp.org/std/meetings-and-participation/upcoming-meetings>

>
> None

>
> 3. Any other business

>
> New reflector

>
> <http://lists.isocpp.org/mailman/listinfo.cgi/sg19>

>
> Old Reflector

> <https://groups.google.com/a/isocpp.org/forum/#!newtopic/sg19>

> <<https://groups.google.com/a/isocpp.org/forum/?fromgroups=#!forum/sg14>>

>
> Code and proposal Staging area

>
> 4. Review

>
> 4.1 Review and approve resolutions and issues [e.g., changes to SG's
> working draft]

>
> 4.2 Review action items (5 min)

>
> 5. Closing process
>
> 5.1 Establish next agenda
>
>
> 5.2 Future meeting
> * Jan 11, 2024 02:00 PM ET: Graph DONE
> * Feb 8, 2024 02:00 PM ET: Graph DONE
> * Mar 14, 2024 02:00 PM ET: Cancelled due to Tokyo 3-18-23
> * Apr 11, 2024 02:00 PM ET: Stats/Graph DONE
> * May 9, 2024 02:00 PM ET: Graph DONE
> * June 13, 2024 02:00 PM ET: Graph; St.louis 6-24-29 DONE
> * July 11, 2024 02:00 PM ET: Stats
> * Aug 15, 2024 02:00 PM ET: Graph
> * Sep 12, 2024 02:00 PM ET: CPPCON Sept 15-20 so cancelled
> * Oct 10, 2024 02:00 PM ET: Stats
> * Nov 14, 2024 02:00 PM ET: Wroclaw F2F
> * Dec 12, 2024 02:00 PM ET: Graph
>
> On Tue, Nov 12, 2024 at 11:54 AM Oliver Rosten via SG19 <
> sg19_at_[hidden]> wrote:
>
>> I'm happy to take a look at the drafts offline, after Poland.
>>
>> O.
>>
>> On Mon, 11 Nov 2024 at 19:21, Phil Ratzloff via SG19 <
>> sg19_at_[hidden]> wrote:
>>
>>> There are updates to the P3126 Overview and P3127 Background and
>>> Terminology papers for the Graph Library. They haven't been published yet,
>>> but it would be helpful to get input on them before they're published.
>>>
>>>
>>>
>>> Since the changes are a response to Oliver's concerns, I think Oliver
>>> and Andrew would need to attend to make it useful to have a session for the
>>> Graph Library. Unless they want to do that, I think we can send draft
>>> versions to the reflector, unless there's a better idea.
>>>

>>>

>>>

>>>

>>>

>>>

>>>

>>> *From:* SG19 <sg19-bounces_at_[hidden]> *On Behalf Of *Michael

>>> Wong via SG19

>>> *Sent:* Monday, November 11, 2024 1:14 PM

>>> *To:* sg19_at_[hidden]

>>> *Cc:* Michael Wong <fraggamuffin_at_[hidden]>

>>> *Subject:* [isocpp-sg19] SG19 Nov call

>>>

>>>

>>>

>>> *EXTERNAL*

>>>

>>> Hi all,

>>>

>>>

>>>

>>> I will unfortunately be travelling. I am still trying to figure out if I

>>> can manage the call from the opposite timezone around 2 am. Even if I were

>>> to get a volunteer to chair in my absence, I would still need to start the

>>> Zoom call which can only be done from my account.

>>>

>>>

>>>

>>> But first thing first, any subject matters and any volunteer chair?

>>> Thank you. Regards

>>> --

>>> SG19 mailing list

>>> SG19_at_[hidden]

>>> <https://lists.isocpp.org/mailman/listinfo.cgi/sg19>

>>>

>> --

>> SG19 mailing list

>> SG19_at_[hidden]

>> <https://lists.isocpp.org/mailman/listinfo.cgi/sg19>

>>

>

Minutes for 2024/12/12 SG19 Conference Call

On Thu, Dec 12, 2024 at 11:13 AM Michael Wong <fraggamuffin_at_[hidden]> wrote:

- > Hi, this SG19 meeting will focus on stats and graphs.
- >
- > Michael Wong is inviting you to a scheduled Zoom meeting.
- >
- > Topic: SG19 monthly
- > Time: 2nd Thursdays 02:00 PM Eastern Time (US and Canada)
- > Every month on the Second Thu,
- >
- >
- > Join from PC, Mac, Linux, iOS or Android:
- >
- > <https://iso.zoom.us/j/93084591725?pwd=K3QxZjJlcnljaE13ZWU5cTILNkx0Zz09>
- > Password: 035530
- >
- > Or iPhone one-tap :
- > US: +13017158592,,93084591725# or +13126266799,,93084591725#
- > Or Telephone:
- > Dial(for higher quality, dial a number based on your current location):
- > US: +1 301 715 8592 or +1 312 626 6799 or +1 346 248 7799 or +1
- > 408 638 0968 or +1 646 876 9923 or +1 669 900 6833 or +1 253 215 8782
- > or 877 853 5247 (Toll Free)
- > Meeting ID: 930 8459 1725
- > Password: 035530
- > International numbers available: <https://iso.zoom.us/u/agewu4X97>
- >
- > Or Skype for Business (Lync):
- > <https://iso.zoom.us/skype/93084591725>
- >
- > Agenda:
- >
- > 1. Opening and introductions
- >
- > The ISO Code of conduct:
- > <https://www.iso.org/files/live/sites/isoorg/files/store/en/PUB100397.pdf>
- >
- > IEC Code of Conduct:

- >
- > <https://www.iec.ch/basecamp/iec-code-conduct-technical-work>
- >
- > ISO patent policy.
- >
- >
- >
- > [https://isotc.iso.org/livelink/livelink/fetch/2000/2122/3770791/Common_Policy.htm?no
deid=6344764&vernum=-2](https://isotc.iso.org/livelink/livelink/fetch/2000/2122/3770791/Common_Policy.htm?no
deid=6344764&vernum=-2)
- >
- > The WG21 Practices and Procedures and Code of Conduct:
- >
- > <https://isocpp.org/std/standing-documents/sd-4-wg21-practices-and-procedures>
- >
- > 1.1 Roll call of participants
- >
- Michael, Phil, Richard, Scott , ozan, Pete
- >
- >
- > 1.2 Adopt agenda
- >
- > 1.3 Approve minutes from previous meeting, and approve publishing
- > previously approved minutes to ISOCPP.org
- >
- > 1.4 Action items from previous meetings
- >
- > 2. Main issues (125 min)
- >
- > 2.1 General logistics
- >
- >
- >
- > * Jan 11, 2024 02:00 PM ET: Graph DONE
- > * Feb 8, 2024 02:00 PM ET: Graph DONE
- > * Mar 14, 2024 02:00 PM ET: Cancelled due to Tokyo 3-18-23
- > * Apr 11, 2024 02:00 PM ET: Stats/Graph DONE
- > * May 9, 2024 02:00 PM ET: Graph DONE
- > * June 13, 2024 02:00 PM ET: Graph; St.louis 6-24-29 DONE
- > * July 11, 2024 02:00 PM ET: Stats/ Graphs DONE
- > * Aug 15, 2024 02:00 PM ET: Graph DONE

> * Sep 12, 2024 02:00 PM ET: CPPCON Sept 15-20 so canceled DONE
> * Oct 10, 2024 02:00 PM ET: Stats DONE
> * Nov 14, 2024 02:00 PM ET: Wroclaw F2F DONE
> * Dec 12, 2024 02:00 PM ET: Graph
>
>
> ISO meeting status
>
> future C++ Std meetings
>
1. Contracts was voted in EWG
2. Wiki password changed
3. Safety concerns
<https://thenewstack.io/feds-critical-software-must-drop-c-c-by-2026-or-face-risk/>

>
>
> 2.2 Paper reviews
>
>

Action Items

- [] Send another email to the LWG chair to ask if they want to see the updated statistics library proposal again.
- [] Make the changes discussed in the meeting to the statistics library proposal before sending the email.

Updates on Graph Library and Benchmarking

- Phil provides an update on the graph library, mentioning changes to containers and views.
- The focus on benchmarking between BGL and the new graph library is discussed, with the need for more data to create the paper.
- The prototype library is being integrated into the company for feedback, with progress being slower than expected.

Discussion on Statistics Library and Implementation Details

- Richard presents updates on the statistics library, discussing the inclusion of mean, variance, skewness, and kurtosis.

- The decision to drop weighted versions of skewness and kurtosis due to conflicting definitions is mentioned.
- The importance of clear documentation and the potential for future amendments on parallel implementation are discussed.
- The need for overloaded constructors and functions to allow for type promotion and user control is highlighted.

Finalizing the Paper and Next Steps

- Richard discusses the need for additional overloads and the potential for parallel implementation to return a reference to the accumulator object.
- The importance of clear documentation and the potential for future amendments on parallel implementation are discussed.
- The need for overloaded constructors and functions to allow for type promotion and user control is highlighted.
- plans to send an email to LWG for further review.

Closing Remarks and Holiday Wishes

- Michael Wong thanks everyone for their contributions and outlines the next steps for the statistics library.
- The importance of targeting C++29 and the opportunities in Bulgaria are discussed.
- The need for additional changes and the potential for future amendments on parallel implementation are highlighted.
- The meeting concludes with holiday wishes and plans to reconvene in January.

> P3495 from Oliver/Mark

>

> >From Phil: There are updates to the P3126 Overview and P3127 Background
> and

> Terminology papers for the Graph Library. They haven't been published yet,
> but it would be helpful to get input on them before they're published.

>

>

>

> Since the changes are a response to Oliver's concerns, I think Oliver and
> Andrew would need to attend to make it useful to have a session for the

- > Graph Library. Unless they want to do that, I think we can send draft
- > versions to the reflector, unless there's a better idea.
- >
- >
- > Review BSI Graph feedback:
- > As Oliver (Rosten) said "The basic premise is important, and it would be
- > fantastic to have support for graphs in the standard."
- >
- > The main items identified were:
- > Oliver:
- > - This paper is long and incomplete, it has lots of details which I think
- > to be irrelevant, however things that are definitely relevant are missing
- > from the paper - for example definition of graph - since people have
- > different ideas. We need to add a mathematical perspective to the paper.
- >
- > - The structure of the paper completely changed in the new revision, so now
- > it's hard to understand what and why they have done
- >
- > - Another missing part is discussion of graph invariants
- >
- > Tom (Deakin): There's a big missing part in "Prior art" part, GraphBLAS (
- > <https://graphblas.org>) eminently.
- >
- > Some other things to add:
- >
- > 1. The electrical circuit example needs more explanation, and I think this
- > will highlight some deep issues around representing things which are
- > seemingly trivially graphs, as graphs in practice. In what sense is a
- > bog-standard resistor directed? I assume the reason that the graph is
- > directed is because current has a sign and in an undirected graph it
- > becomes ambiguous which way the current is flowing (also you may want
- > components like diodes). But the directed representation also has issues:
- > "can current flow from 'Vdd' to 'n0'?" should be immediately answerable
- > from the properties of Vdd and its edges. There are other ways to represent
- > an electrical circuit. One is as a directed graph but with incident edges
- > recorded - but iirc, this is excluded from the latest version of the paper.
- > Alternatively, one could have a mathematical object, the name of which I
- > actually don't know: it looks like an undirected graph, but where each
- > partial edge has additional, unique, end-point data, as well as the common
- > weight. Things like this are the reason why I think we need a broader group
- > to look at this proposal (i.e. beyond SG19) and if we possibly can we

> should involve someone from the mathematics community. Otherwise there's a
> real danger we end up missing important insights.

>

> 2. My comment about the structure of the paper changing was a reference to
> previous comparisons with `boost::graph`. I'm sure these were in an earlier
> version, or am I misremembering? Either way, it would be very helpful to
> have a proper discussion of e.g. the move away from visitors.

>

> 3. Re. the definition of a graph, there needs to be a proper discussion
> about whether the paper's definition of graph is what some authors call a
> multigraph and whether it does/does not include loops. These things are
> mentioned, in passing, when introducing algorithms, but terminology needs
> to be properly established.

>

> 4. I think we're trying to do too much in one go in this paper. I think a
> great first step would be to build on `mdspan` and try to standardize (or at
> least understand) what might reasonably be called an unstructured span.
> This could be represented as a vector of vectors or as a vector with some
> auxiliary storage indicating where the partitions fall. The point is that
> an unstructured span, with the right invariants, is an adjacency list. If
> we can understand unstructured span and its desirable api, I think this
> will be incredibly valuable guidance for what a standardized graph
> container might look like.

>

> 5. IIUC, this paper excludes pure connectivity graphs. These are incredibly
> helpful and, if I've understood correctly that they are not supported,
> would be a major omission. Another good reason, imo, to start with
> unstructured span!

>

> 6. I'm not convinced by the load api. We don't have a load api for vector
> etc. Moreover, would it not be preferable to have appropriate constructors?

>

>

> 2.2.1: ML topics

>

> 2.2.1.1 Graph Proposal Phil Ratsloff et al

>

> Latest paper:

>

> Here's a link to the paper (different than the previous paper reviewed).
> There are some additional updates I'm planning on making before the

> meeting.

>

>

>

<https://docs.google.com/document/d/1OpH-xxRri7tJTtJJIZTYmSHkkrZJkdBwm9zJ7LgolfQ/edit?usp=sharing>

>

>

>

>

> P1709R3:

>

>

https://docs.google.com/document/d/1kLHhbSTX7j0tPeTYECQFSNx3R35Mu3xO5_dyYdRy4dM/edit?usp=sharing

>

>

>

https://docs.google.com/document/d/1QkfDzGyfNQKs86y053M0YHOLP6frzhTJqzg1Ug_vkkE/edit?usp=sharing

>

> <<http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2020/p2119r0.html>>

>

> <

>

>

<https://docs.google.com/document/d/175wlm8o4BNGti0WLq8U6uZORegKVjmnpsc-E8PoGS0/edit?ts=5fff27cd#heading=h.9ogkehmdmtel>

> *>*

>

> Array copy semantics:

> array copy-semantics paper P1997 "Relaxing Restrictions on Arrays",

> <https://wg21.link/p1997>

>

> Stats feedback:

>

> P2376R0

> <<http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2021/p2376r0.pdf>>

> Comments

> on Simple Statistical Functions (p1708r4): Contracts, Exceptions and

> Special cases Johan Lundberg

- >
- > 2.2.1.2 Reinforcement Learning Larry Lewis Jorge Silva
- >
- > Reinforcement Learning proposal:
- >
- > 2.2.1.3 Differential Calculus:
- >
- >
- >
- <https://docs.google.com/document/d/175wlm8o4BNGti0WLq8U6uZORegKVjmnpc-E8PoGS0/edit?ts=5fff27cd#heading=h.9ogkehmdmtel>
- >
- > 2.2.1.4: Stats paper
- >
- > P2681R0
- > <<https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2022/p2681r0.pdf>>
- > More
- > Stats Functions Richard Dosselmann, Michael Wong
- > Current github
- >
- > <https://github.com/cplusplus/papers/issues/475>
- >
- > <https://github.com/cplusplus/papers/issues/979>
- >
- > Stats review Richard Dosselman et al
- >
- > <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2021/p1708r4.pdf>
- >
- > Feedback from Johan Lundberg and Oleksandr Korval
- >
- > <https://isocpp.org/files/papers/D2376R0.pdf>
- >
- > P1708R3: Math proposal for Machine Learning: 3rd review
- >
- > PXXXX: combinatorics: 1st Review
- >
- > *> std.org/jtc1/sc22/wg21/docs/papers/2020/p1708r2
- > <<http://std.org/jtc1/sc22/wg21/docs/papers/2020/p1708r2>>*
- > *> above is the stats paper that was reviewed in Prague*
- > *> <http://wiki.edg.com/bin/view/Wg21prague/P1708R2SG19>
- > <<http://wiki.edg.com/bin/view/Wg21prague/P1708R2SG19>>*

> *>*

> *> Review Jolanta Polish feedback.*

> *> <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2020/p2119r0.html>

> <<http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2020/p2119r0.html>>*

>

>

> 2.2.1.4: Matrix paper

>

> 2.2.3 any other proposal for reviews?

>

> 2.3 Other Papers and proposals

>

> P1416R1: SG19 - Linear Algebra for Data Science and Machine Learning

>

>

<https://docs.google.com/document/d/1IKUNiUhBgRURW-UkspK7fAAyIhfXuMxjk7xKikK4Yp8/edit#heading=h.tj9hitg7dbtr>

>

> P1415: Machine Learning Layered list

>

>

https://docs.google.com/document/d/1eINFdIXWoetbxjO1OKol_Wj8fyi4Z4hogfj5tLVSi64/edit#heading=h.tj9hitg7dbtr

>

> 2.2.2 SG14 Linear Algebra progress:

> Different layers of proposal

>

>

https://docs.google.com/document/d/1poXfr7mUPovJC9ZQ5SDVM_1Nb6oYAXIK_d0ljdUAtSQ/edit

>

> 2.5 Future F2F meetings:

>

> 2.6 future C++ Standard meetings:

> <https://isocpp.org/std/meetings-and-participation/upcoming-meetings>

>

> None

>

> 3. Any other business

>

> New reflector

- >
- > <http://lists.isocpp.org/mailman/listinfo.cgi/sg19>
- >
- > Old Reflector
- > <https://groups.google.com/a/isocpp.org/forum/#!newtopic/sg19>
- > <<https://groups.google.com/a/isocpp.org/forum/?fromgroups=#!forum/sg14>>
- >
- > Code and proposal Staging area
- >
- > 4. Review
- >
- > 4.1 Review and approve resolutions and issues [e.g., changes to SG's
> working draft]
- >
- > 4.2 Review action items (5 min)
- >
- > 5. Closing process
- >
- > 5.1 Establish next agenda
- >
- >
- > 5.2 Future meeting
- > * Jan 11, 2024 02:00 PM ET: Graph DONE
- > * Feb 8, 2024 02:00 PM ET: Graph DONE
- > * Mar 14, 2024 02:00 PM ET: Cancelled due to Tokyo 3-18-23
- > * Apr 11, 2024 02:00 PM ET: Stats/Graph DONE
- > * May 9, 2024 02:00 PM ET: Graph DONE
- > * June 13, 2024 02:00 PM ET: Graph; St.louis 6-24-29 DONE
- > * July 11, 2024 02:00 PM ET: Stats
- > * Aug 15, 2024 02:00 PM ET: Graph
- > * Sep 12, 2024 02:00 PM ET: CPPCON Sept 15-20 so cancelled
- > * Oct 10, 2024 02:00 PM ET: Stats DONE
- > * Nov 14, 2024 02:00 PM ET: Wroclaw F2F DONE
- > * Dec 12, 2024 02:00 PM ET: Graph